

1.

a) List three features of every object-oriented programming language.

- Encapsulation
- Inheritance
- Polymorphism

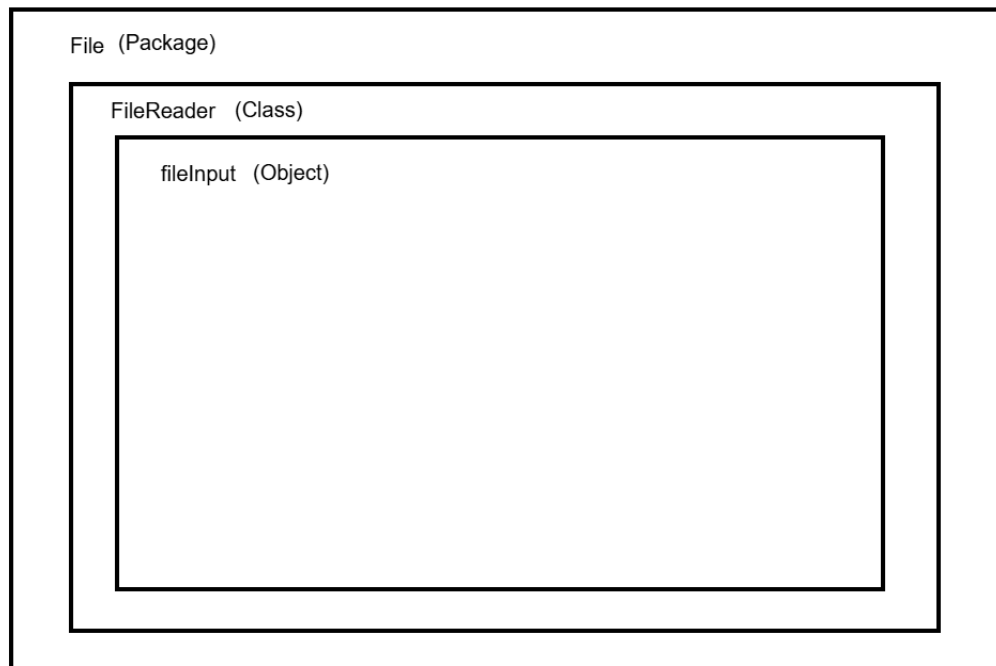
b) Explain, in your own words, what you think the meaning is of each of the three features listed in part (a).

Encapsulation: Bundling data and functions into a single unit. Restricts direct interaction with certain information or functions, also known as information hiding. Prevents external code from modifying an object's state which ensures data integrity.

Inheritance: A unit inheriting data from their parent class. Eliminates redundant code and establishes a class hierarchy.

Polymorphism: Allows subclasses to be treated the same. Creates highly flexible code.

2. Draw a sketch that shows the relationship, and/ or hierarchy, of object, class, and package for the package File, class FileReader, and the object fileInput.



3. Write an appropriate comment block for the beginning of a program to describe an application that is intended to calculate test averages.

```
/**  
Calculates test averages  
*/
```

4. Write an appropriate comment for describing a class that displays the average of a set of scores.

```
/**  
Displays the average of a set of scores  
*/
```

5. Write a statement that declares a package with the name gradeCalculator.

```
package gradeCalculator;
```

11.

1. The comment indicator has to be altogether in the same line
2. the main function should be public, not private
3. the print statement needs a closing semicolon